Internal Flow and Heat Transfer in a Centrifugal Pump

Experiment 3 | 20ME406P Computational Fluid Dynamics and Heat Transfer Lab

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1. Aim and Scope

To study the centrifugal pressure rise, the absolute and relative velocities, the pump head and the temperature change in a simplified rotating annular passage, and to check how the pump head scales with rotational speed.

- The run analysed in this report is a user-defined (custom) case at 3000 rpm and Re = 500, on a 100 × 200 polar grid. The compulsory assignment (1000, 2000 and 3000 rpm at fixed Reynolds number) is used to assess the expected head scaling $H \propto RPM^2$.
- The solver is a reduced-order teaching model with prescribed velocity components. It is not blade-resolved pump CFD and is not suitable for predicting efficiency maps, so the results are presented as numerical trends and not as validated pump performance data.

2. Governing Equations

2.1 Centrifugal pressure rise and pump head

In the single rotating frame the impeller rotates at N rpm, which corresponds to the angular speed

$$\Omega = \frac{2\pi N}{60} \tag{1}$$

The pressure along the radius of the annular passage (hub radius r1, shroud radius r2, channel depth b) is the centrifugal contribution minus a small viscous loss for the radial through-flow with inlet radial velocity U1:

$$p(r) = \frac{1}{2}\rho\Omega^2\left(r^2 - r_1^2\right) - \frac{12\mu U_1 r_1}{b^2}\ln\frac{r}{r_1} \tag{2}$$

The leading-order pressure rise between the hub and the shroud, and the corresponding pump head, are therefore

$$\Delta p = p(r_2) - p(r_1) \approx \frac{1}{2}\rho\Omega^2\left(r_2^2 - r_1^2\right) \tag{3}$$

$$H = \frac{\Delta p}{\rho g} \tag{4}$$

2.2 Velocity field

The velocity field is prescribed and not solved from a momentum equation. The radial velocity follows from continuity for radial through-flow in a constant-depth annulus, with the inlet velocity set by the Reynolds number based on the hydraulic diameter:

$$u_r = \frac{U_1 r_1}{r}, \qquad U_1 = \frac{Re\,\mu}{\rho D_h}, \qquad D_h = 2b \tag{5}$$

$$Re = \frac{\rho U_1 D_h}{\mu} \tag{6}$$

The absolute tangential velocity is a fixed fraction σ (the swirl factor) of the blade speed Ωr. The tangential velocity relative to the rotating frame is the absolute value minus Ωr:

$$u_{\theta,abs} = \sigma \Omega r, \qquad u_{\theta,rel} = u_{\theta,abs} - \Omega r = (\sigma - 1)\Omega r \tag{7}$$

2.3 Thermal transport

Temperature is transported by the relative velocity in the rotating frame, with conduction and a source term ST:

$$\rho c_p (\mathbf{u}_{rel} \cdot \nabla T) = k \nabla^2 T + S_T \tag{8}$$

The source term represents heat exchange with the walls (heat-transfer coefficient h at wall temperature Tw, applied on both faces of the channel) and the heating caused by hydraulic losses, which is taken as a fraction ε of the rotational power term:

$$S_T = \frac{2h}{b}(T_w - T) + \rho\, \varepsilon\, \Omega^2 r\, u_r \tag{9}$$

2.4 Derived quantities

The mass flow rate through the inlet circle and the heat gained by the fluid are

$$\dot{m} = \rho U_1\, 2\pi r_1 b, \qquad Q = \dot{m}\, c_p (T_{out} - T_{in}) \tag{10}$$

Since Eq. (3) is proportional to Ω^2, the head is expected to follow the pump-affinity scaling, where k is a constant at fixed Reynolds number:

$$H = k\left(\frac{N}{1000}\right)^2 \tag{11}$$

3. Boundary Conditions

- **Inlet (hub, r = r1):** specified inlet temperature Tin = 300 K, and radial inflow at the eye with velocity U1 from Eq. (5).
- **Outlet (shroud, r = r2):** zero-gradient temperature, $\partial T/\partial r = 0$, with outward radial transport.
- **Azimuthal direction:** periodic in θ, so the first and last angular nodes are neighbours.
- **Walls:** heat exchange is not applied as a surface boundary condition but is represented through the distributed source term in Eq. (9), with Tw = 330 K and h = 120 W/m²K.
- **Pressure:** not solved. It is evaluated from the analytic Eq. (2), which is zero at the hub and increases with radius.
- **Initial condition:** uniform temperature T = Tin everywhere in the annulus.

4. Numerical Method

4.1 Grid and discretisation

A polar single-rotating-frame model is used on a structured grid of nr nodes in the radial direction (including both hub and shroud) and nθ equally spaced nodes in the azimuthal direction (the closing node at 2π is not duplicated). Velocities and pressure are evaluated analytically at the grid nodes from Eqs. (2) and (5)-(7). The heat equation is solved with a finite-difference scheme using first-order upwind convection and central diffusion:

- radial convection is upwinded from the inner neighbour (the radial velocity is positive);
- azimuthal convection is upwinded according to the sign of the relative tangential velocity, which is negative here (the fluid lags the impeller), so the value is taken from the $\theta + \Delta\theta$ neighbour;
- diffusion uses central differences in r and θ, and the wall exchange and loss heating are added as source terms. The radial curvature term $(1/r)\partial T/\partial r$ of the Laplacian is not included in the discrete diffusion operator, which is acceptable here because the thermal diffusivity (1.4×10^{-7} m²/s) is very small.

4.2 Solution algorithm

1. Compute the angular speed, inlet velocity, velocity components, pressure and loss-heating source at all nodes.
2. Initialise the temperature to Tin.
3. In each iteration, compute the new temperature at every node from its neighbours and the sources, and under-relax it with factor 0.85: Tnew = 0.15 Told + 0.85 Tcandidate.
4. Apply the boundary conditions (hub temperature fixed at Tin, shroud zero-gradient) and clip the temperature to within 100 K of the inlet and wall temperatures as a safety bound.
5. Compute the L∞ residual, defined as the maximum absolute temperature change in the iteration, and store the minimum, maximum and mean outlet temperature.
6. Stop when the residual falls below 10^{-10} (after at least 50 iterations) or when the specified number of iterations is reached.

After the iterations, the pressure rise, the pump head, the outlet temperature, the mass flow rate and the fluid heat gain are computed from Eqs. (3), (4) and (10), and the fields are written to VTK files.

4.3 Case definition and parameters

The run analysed in this report corresponds to the following parameter set (RUN_MODE = custom):

Parameter	Symbol	Value
Radial grid points	nr	100
Azimuthal grid points	ntheta	200
Rotational speed	N	3000 rpm (Ω = 314.16 rad/s)

Parameter	Symbol	Value
Reynolds number (through-flow)	Re	500
Inlet temperature	Tin	300 K
Wall temperature	Tw	330 K
Iterations	iterations	1200
Swirl factor	σ	0.85
Wall heat-transfer coefficient	h	120 W/m²K
Loss fraction	ε	0.04
Hub and shroud radius	r1, r2	0.030 m, 0.120 m
Channel depth	b	0.010 m (D_h = 0.020 m)
Water properties	ρ, μ, cp, k	997 kg/m³, 1.0×10^{-3} Pa·s, 4180 J/kgK, 0.60 W/mK
Inlet radial velocity	U1	0.02508 m/s (0.00627 m/s at shroud)
Mass flow rate	ṁ	0.0471 kg/s
Under-relaxation factor	-	0.85

4.4 Compulsory assignment cases

The pump head was evaluated at 1000, 2000 and 3000 rpm with the Reynolds number fixed at 500, so that the inlet velocity U1 is the same in all three cases and only the rotational speed changes. The head is then plotted against $(RPM/1000)^2$ as in Eq. (11).

Model scope: reduced-order annular model with prescribed velocities and an analytic pressure field. No blades, impeller-casing interaction, slip or secondary flows are resolved.

5. Simulation Outputs

The figures below are the outputs of the Colab runs and are discussed.

5.1 Colab parameter cell

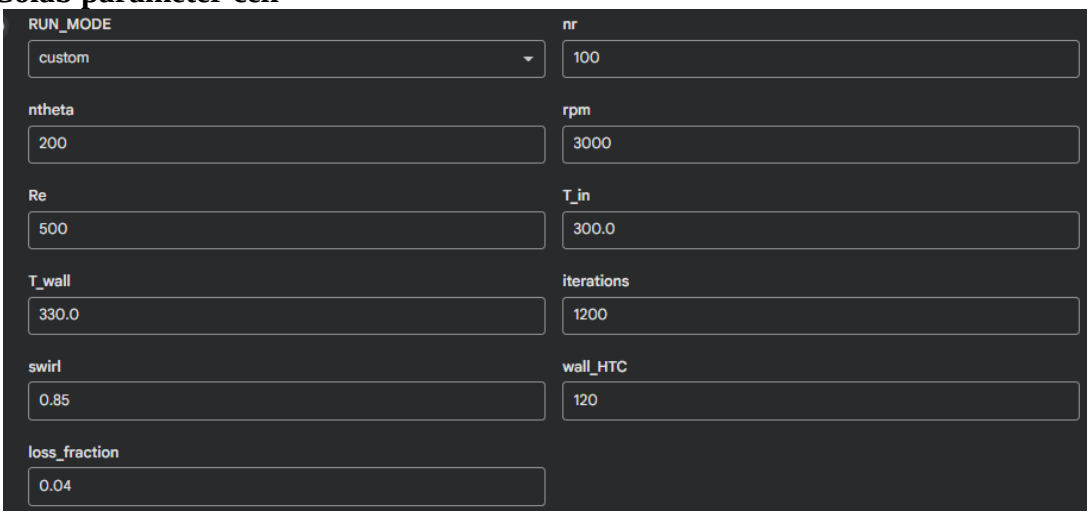


Figure 1. Run mode and parameter set used in the Colab notebook (RUN_MODE = custom, nr = 100, ntheta = 200, rpm = 3000, Re = 500, Tin = 300 K, Twall = 330 K, iterations = 1200, swirl = 0.85, wall_HTC = 120, loss_fraction = 0.04).

5.2 Thermal convergence and outlet temperature

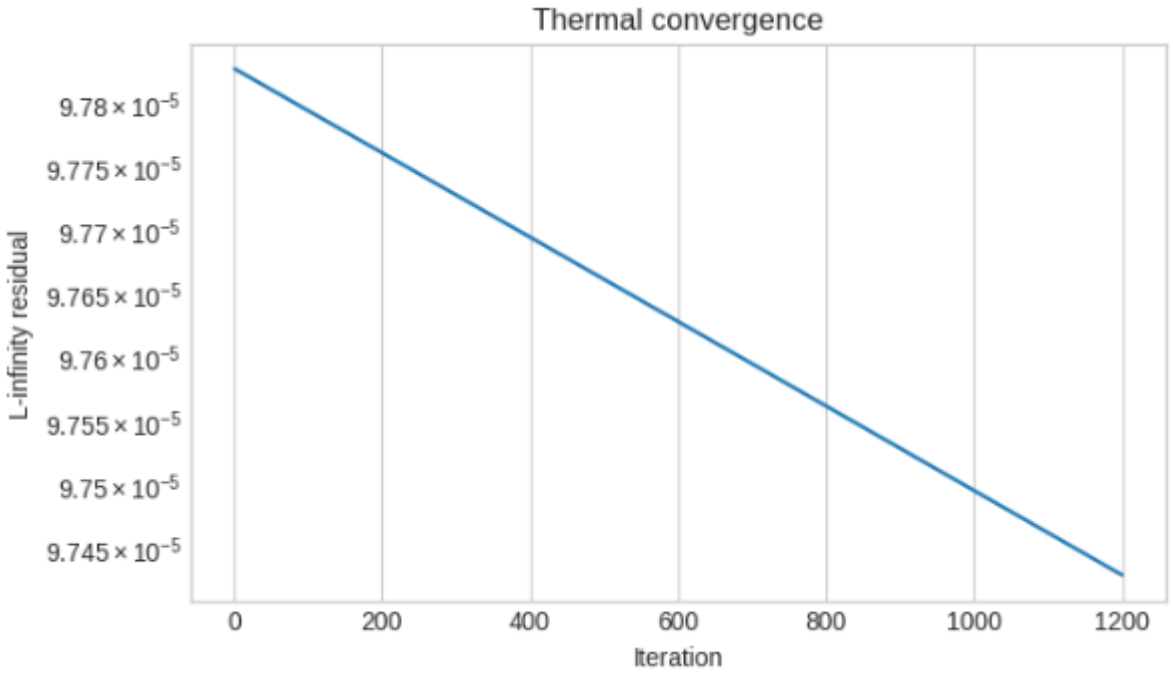


Figure 2. Thermal convergence: L-infinity residual versus iteration.

- In Figure 2 the L∞ residual starts at 9.78×10^{-5} K and ends at 9.74×10^{-5} K after 1200 iterations, a fall of only about 0.4 %. It does not decay towards the stopping tolerance of 10^{-10}, so the solver ran the full 1200 iterations without meeting its convergence criterion.

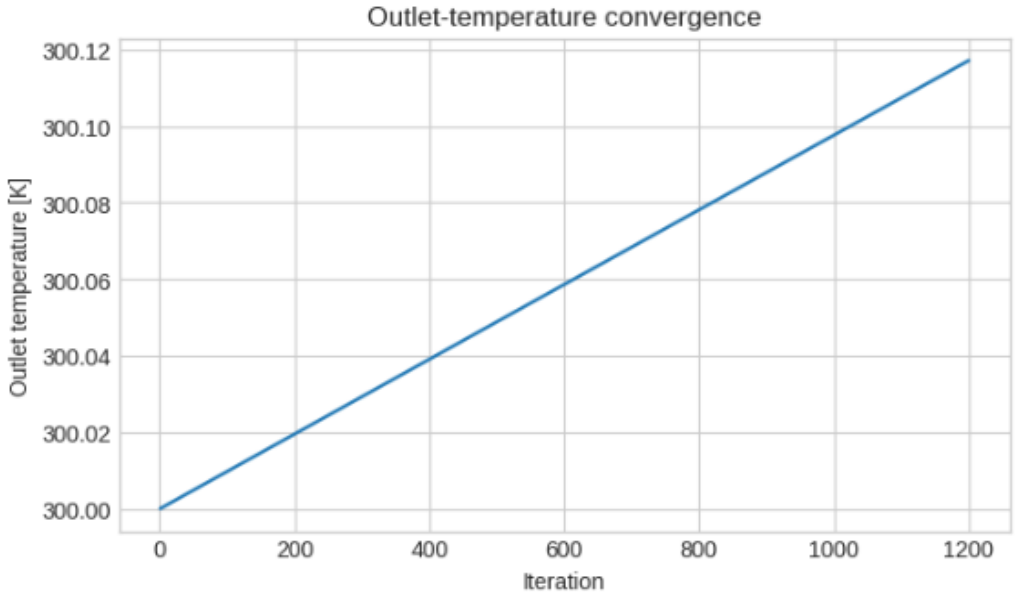


Figure 3. Outlet temperature versus iteration.

- Figure 3 shows the outlet temperature increasing almost perfectly linearly from 300.000 K to 300.117 K, which is about 9.8×10^{-5} K per iteration. This equals the residual, because the outlet is the hottest region and therefore carries the largest change in each iteration.
- A straight rising line together with an almost constant residual shows that the temperature field is still changing at a constant rate and has not reached a steady state. The outlet temperature rise of 0.117 K and the corresponding fluid heat gain of about 23 Ware therefore intermediate values and not converged results.
- A likely reason is that the through-flow velocity is very small (0.025 m/s at the hub, falling to 0.006 m/s at the shroud), so the pseudo-time iteration advances the thermal field very slowly.
- Many more iterations, or an implicit solver, would be needed to obtain a steady thermal solution. The minimum temperature remains 300.0 K at the inlet and the maximum occurs at the outlet, so the fluid is heated in the direction of flow, as expected for a 330 K wall.

5.3 Pressure field

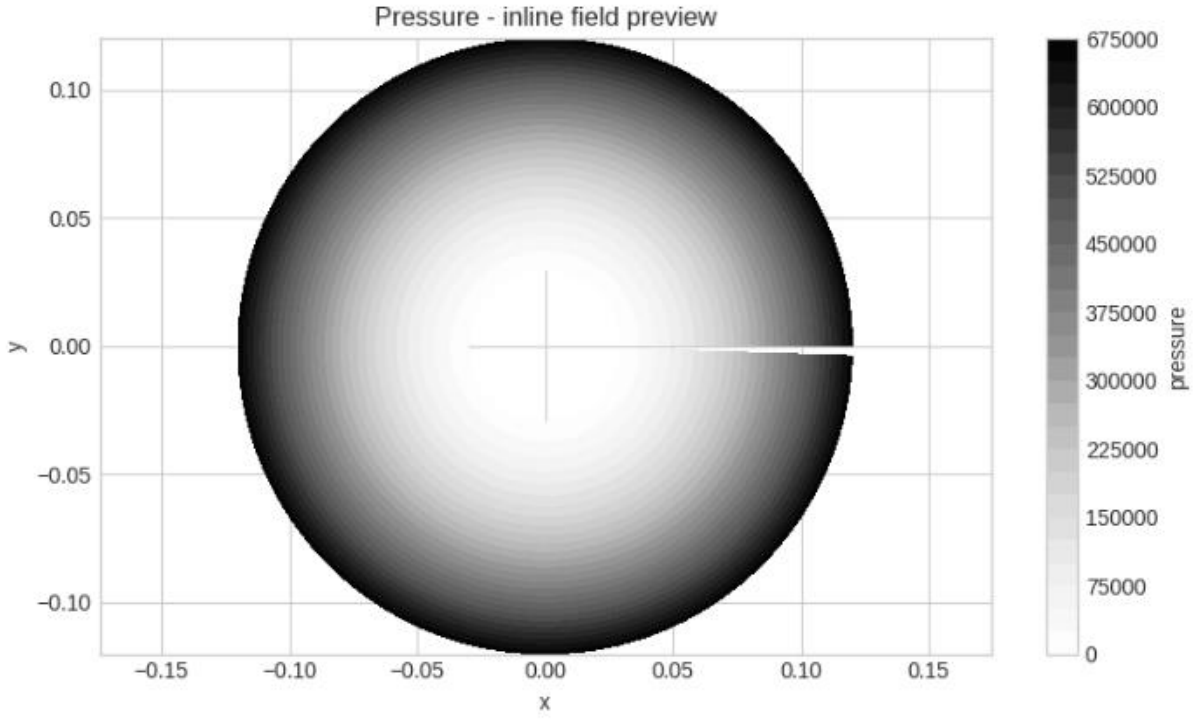


Figure 4. Pressure field in the annular passage at 3000 rpm (inline preview of pump_annulus_srf.vtk).

- The pressure field at 3000 rpm in Figure 4 consists of concentric circles. The pressure is zero at the hub (r = 0.03 m) and increases with radius up to about 664 kPa at the shroud (r = 0.12 m), which is consistent with the colour-bar limit of 675 kPa. The increase is quadratic in r, so it is gentle near the hub and steep near the shroud.

- There is no variation with angle because the model prescribes an axisymmetric field, so no blade-to-blade pressure difference is predicted.
- The thin white wedge on the right-hand side near $\theta = 0$ is a plotting artefact: the azimuthal grid stops before 2π and the contour plot therefore does not close. It is not a physical feature.
- The viscous loss term in Eq. (2) contributes only about 0.125 Pa, compared with 664 kPa from the centrifugal term, so the pressure rise is essentially the centrifugal value of Eq. (3). At 2000 rpm (Figure 5, right) the shroud pressure is about 295 kPa, which equals the 3000 rpm value multiplied by $(2/3)^2$.

5.4 Pump head scaling with RPM²

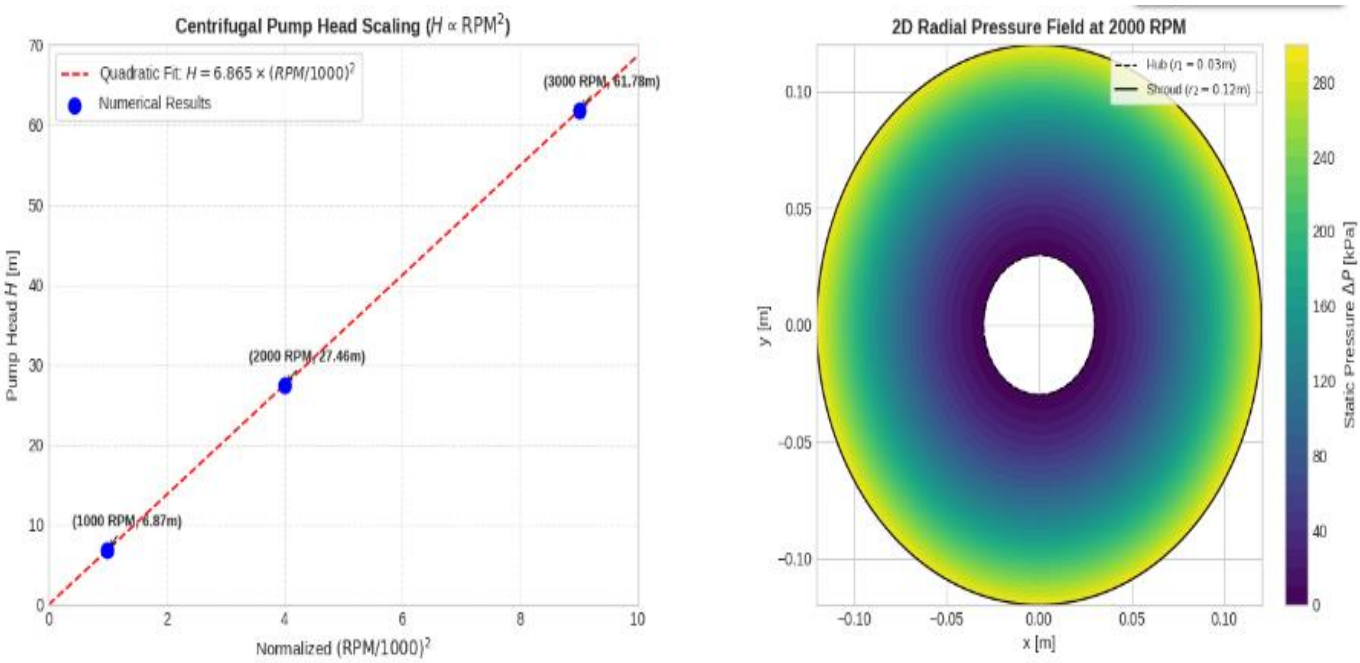


Figure 5. Left: pump head against $(RPM/1000)^2$ with quadratic fit. Right: radial pressure field at 2000 rpm.

- Figure 5 (left) shows that the three points lie on a straight line through the origin when the head is plotted against $(RPM/1000)^2$.
- The head per unit $(RPM/1000)^2$ is the same at 1000, 2000 and 3000 rpm, so doubling the speed gives four times the head and tripling it gives nine times. This is the expected scaling $H \propto N^2$ of Eq. (11), and it follows directly from Eq. (3), because Δp is proportional to Ω^2.
- The Reynolds number is fixed, so the viscous term is the same at all speeds and is too small to cause any visible departure from the quadratic law (relative deviation of the order of 10^{-7}).

6. Results and Discussion

Numerical values quoted from the figures are read from the plotted curves and are therefore approximate. Highlighted values should be checked against the solver output files before submission.

RPM	Ω (rad/s)	Δp (kPa)	H (m)	H / (RPM/1000)² (m)
1000	104.72	73.80	6.87	7.55
2000	209.44	295.20	27.46	7.55
3000	314.16	664.20	61.78	7.55

- This scaling is an exact consequence of the analytic pressure model, so it verifies the implementation and the pump-affinity law, but it is not an independent prediction by a flow solution. The pressure rise in the model is also an ideal upper bound.
- It uses the full rigid-body value $\frac{1}{2}\rho\Omega^2(r2^2 - r1^2)$ and does not contain the swirl factor, whereas a real impeller loses head through slip, blade loading and hydraulic losses.

6.1 Velocities and quantitative summary

Quantity (3000 rpm run)	Value
Angular speed Ω	314.16 rad/s
Shroud pressure rise Δp	≈ 664 kPa
Pump head H	≈ 67.9 m
Absolute tangential speed (hub → shroud)	8.0 → 32.0 m/s
Relative tangential speed (hub → shroud)	−1.4 → −5.65 m/s
Radial velocity (hub → shroud)	0.0251 → 0.0063 m/s
Maximum absolute speed (at shroud)	≈ 32.0 m/s
Inlet / outlet temperature	300.000 K / 300.117 K
Fluid heat gain (not converged)	≈ 23 W
Final L∞ residual	9.74×10^{-5} K

- The absolute tangential velocity is more than three orders of magnitude larger than the radial velocity at the shroud (32.0 m/s against 0.006 m/s), so the flow is dominated by swirl.
- The relative tangential velocity is negative, meaning the fluid lags behind the impeller by 15 % of the blade speed in this model, and it reaches about 5.65 m/s at the shroud. Because the Reynolds number is based on the through-flow, it does not change with rpm, and the rotational Reynolds number is much larger (about 7.5×10^5 based on the tip speed and the hydraulic diameter).

6.2 Limitations

- The velocity field and the pressure are prescribed and analytic, so no momentum equation, blade passage or three-dimensional effect is solved.
- The thermal field has not converged after 1200 iterations, so the temperature results are qualitative.
- The pressure rise uses full rigid-body rotation and does not include slip, swirl effects or hydraulic losses, so the head is an ideal value.
- The walls are at a uniform temperature, the loss heating uses a fixed fraction, and the water properties are constant.

7. Conclusions

- The reduced-order polar model gives a smooth axisymmetric pressure field, rising quadratically from zero at the hub to about 664 kPa at the shroud at 3000 rpm.
- The pump head scales as RPM^2: the head divided by $(RPM/1000)^2$ is constant at 1000, 2000 and 3000 rpm, in line with the pump-affinity law. This follows from the analytic centrifugal term, since the viscous loss is negligible (about 0.125 Pa).
- The flow is swirl-dominated, with absolute tangential speeds up to about 32 m/s and relative tangential speeds up to about 5.65 m/s at the shroud, compared with a through-flow of only 0.006-0.025 m/s.
- The thermal solution is not converged: the residual stays at about 9.7×10^{-5} K and the outlet temperature is still rising linearly (300.000 K to 300.117 K), so the heat gain of about 23 W is an intermediate value. More iterations or an implicit solver are needed for a steady result.
- The results are trend indicators from a reduced teaching model without blades. They should not be used to predict pump efficiency or real head.